From Published Evidence to Analyzable Biomedical Cohorts with Scientific Agents

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Abstract

Biomedical papers often report enough information to motivate a new analysis without releasing the combination of individual measurements it requires. We study whether a scientific agent can turn these findings into complete virtual cohorts for a specified population and analysis. The agent distinguishes numerical constraints from contextual findings, derives parameters where studies use different statistical definitions, and supplies an explicit joint model for relationships the evidence leaves open. We evaluate the generated records by running the requested analyses. In two glycemia cases, adding published associations reduced mean absolute discrepancy for five of six regression coefficients. In an iron-status case, two generators reproduced the source odds ratio of 6.65, yet all six generated tables underestimated the target sample odds ratio of 17.79. One generator closely reproduced the proportion with both biomarkers low; the other closely reproduced their continuous rank correlation. In earlier tests with reference results withheld until generation closed, neither case met its prespecified overall criterion. Published evidence can thus support specific uses of a virtual cohort. Which uses survive transfer depends on the information available about the relationship being analyzed.

CCS Concepts

• **Computing methodologies** → **Modeling and simulation**; • **Applied computing** → *Life and medical sciences.*

Keywords

Synthetic cohorts, biomedical evidence, scientific agents, epidemiology

1 Introduction

Suppose a researcher wants to study iron status among young adult women. She can find a paper giving ferritin concentrations within hemoglobin bands and counts of women jointly classified by both thresholds, but she cannot access the paired measurements [33]. She wants to know how often both values are low, how strongly low ferritin is associated with low hemoglobin, and how the two concentrations vary together over their full ranges. With the original measurements, she could answer all three questions on the same rows. With a published table, she can recover the first two quantities for the source sample; she cannot recover the third exactly, or assume that any of them will carry over unchanged to another population.

This is a common use for synthetic biomedical data: the records needed for an analysis are inaccessible, incomplete, or were never measured together. The researcher may have no individual records at all. Studies nevertheless report means, subgroup frequencies and regression coefficients that constrain a possible cohort. The challenge is deciding what each result constrains. A crude odds ratio for raw ferritin and an odds ratio for inflammation-corrected ferritin describe different measurements [33,34]. Even two regression coefficients measured in the same units need not belong in the same multivariable equation. A system that treats these findings as interchangeable can generate plausible-looking records that answer the wrong question.

We give a scientific agent the intended analysis, target population, admissible evidence and any permitted observations. The agent chooses how to use each finding, authors an executable joint probability model and records the assumptions needed to sample from it. Statistical tools then draw complete records and run the researcher's analyses. The output is a dataset that can be analyzed with ordinary software, together with a specification showing which relationships came from observations and which the agent supplied. The research question is visible during construction; a measured target result used for evaluation need not be.

We organize the experiments around three questions. **RQ1:** When the agent receives external relationship evidence in addition to marginal information, which target regression estimates improve? **RQ2:** Which analyses can the same continuous virtual cohort support, including analyses not numerically specified by a source? **RQ3:** How does that pattern change across biomedical requests, and what survives when measured reference results are first revealed after generation? These questions separate agreement with an input statistic, the usefulness of a completed joint model, and transfer to a new reference.

2 Related work and problem definition

2.1 From available information to analyzable records

Filshtein et al. share fitted study models so a recipient can simulate complete cohorts for epidemiological analyses [1]. SynthIPD uses published survival curves and trial summaries to reconstruct time-to-event records with covariates [2]. SynDI combines internal individual data with external regression information across populations to fit a target regression [29]. Multilevel network meta-regression likewise relates aggregate evidence to individual-level models under population assumptions [3]. These approaches differ in what the recipient knows. Filshtein et al. supply the comparison for our HRS experiment because the published exports and the scientific analysis can be matched. In our external-evidence experiments, the agent must also choose the joint dependence structure needed to perform the requested analyses.

G-Sim combines LLM-authored simulators with numerical calibration and structural refinement [4]. LLMSynthor uses aggregate discrepancies to guide proposals for new records and evaluates mobility, e-commerce and household population synthesis [5]. SynthCraft studies healthcare synthetic-data workflows [6], and SPADA samples from a dependency graph fitted with observed tabular data

[15]. Our methodological question concerns the preceding biomedical decision: how findings with different populations, scales and adjustment sets constrain one generator, and which analyses the resulting data support.

2.2 Evidence-conditioned cohort construction

A request specifies the target population, eligibility and measurements, desired variables, permitted observations and intended analyses. Each evidence record identifies its source population, measurement scale, statistical operator, adjustment set, estimate, available uncertainty and provenance. The agent returns an executable generator, the numerical uses of that evidence and the assumptions needed to complete the model:

```
(generator, evidence uses, completion assumptions)

    = agent(scientific request, evidence, optional
     observations)

virtual data = generator(number of rows, random seed)

scientific results = requested analyses(virtual data).
```

In whole-cohort construction, optional observations are empty and all requested variables are generated. In enrichment, supplied covariates remain fixed while an unmeasured variable is generated. These modes share the evidence-to-law problem; the principal experiments here concern complete cohorts.

A finding may constrain the law numerically, provide a check on its implications, or remain contextual. Algebraic consequences of accepted findings are derived evidence. Unreported dependence and transfer from a source population to the request are completion assumptions. Several laws can satisfy the available information while implying different conditional results, so the output includes both the table and its construction specification.

We evaluate usefulness at the level of scientific analysis. For an adjusted coefficient, correlation or dispersion statistic, the generated estimate is compared with the relevant sample reference in its native units. Reproducing a supplied summary measures reconstruction; agreement with a target result not supplied to the author addresses a different question. A table can support several analyses with different degrees of evidential support. These distinctions guide both construction and evaluation.

3 Agent-controlled evidence-to-cohort construction

3.1 Qualify evidence against the scientific request

The workflow begins with the population, era, measurements, units, eligibility and scientific analyses. It accepts curated evidence and agent-directed acquisition. With retrieval enabled, the agent selects queries and primary studies to address evidence gaps; the retained record distinguishes full text, partial access and bibliographic metadata and records overlap or exclusions. Researchers supply the request and access conditions, while the agent chooses how the admitted evidence enters the model.

Qualification preserves each finding's original statistical meaning. Logistic disease coefficients cannot become continuous-biomarker slopes, grouped contrasts cannot become per-unit effects, and rank correlations require different treatment from Pearson correlations. Source and requested adjustment sets must also be reconciled. Unit conversion supplies an algebraic transformation rather than evidence of population or assay equivalence. A finding with unresolved definitions can remain useful context without being promoted to a numerical constraint.

In the external-evidence study, researchers supplied audited source packets and measurement guidance [7–11]. Other implementations combined reviewed articles with optional retrieval or allowed authors to select their own sources through a bounded Europe PMC/PMC gateway. The host dispatched authors and coordinated access, review and execution. These implementations share the scientific decision sequence while preserving their separate evidence budgets and input conditions.

3.2 Translate findings into numerical model constraints

The agent authors a model specification, generator code, evidence-use records and a decision explanation. A relationship counts as numerical evidence use when its role is visible in the implemented law or in an explicit calculation. The common evidence-use record connects relationship, dispersion, scenario, check-only and retained-assumption decisions to sources and specification locations. It distinguishes arithmetic from geometric summaries, marginal from conditional effects, rank from Pearson dependence, and individual variation from uncertainty in an estimate.

A worked example shows why this step requires modeling. Jansen et al. report separate age/sex-adjusted HbA1c projections for BMI and glucose [9]. For residualized BMI and glucose with SDs `sB` and `sG` and correlation `r`, a joint response law with coefficients `thetaB` and `thetaG` implies:

```
cB = thetaB + thetaG * r * sG/sB

cG = thetaG + thetaB * r * sB/sG.
```

CHNS E1 constructor A01, selected by ordinal identifier, used reported slopes `cB = 0.13` and `cG = 2.29`, residual SDs 4.07611 and 0.49694, and the declared assumption `r = 0.25`. Solving the equations gave 0.0642173 and 2.1583154 in IFCC response units. Multiplication by 0.09148 yielded 0.00587460 and 0.19744269 in HbA1c percentage-point units [11]. The source projections therefore informed the joint law through their mathematical relationship, with residual dependence and population transfer remaining explicit assumptions. The worked trace connects the derivation to frozen code, an unfitted correlation check and the resulting target BMI discrepancy.

3.3 Complete an executable joint law

The agent chooses the probability family, dependencies, parameters and remaining completion assumptions. These choices must support complete rows and the requested scientific analyses. Means and SDs alone do not determine residual dependence, tails or subgroup

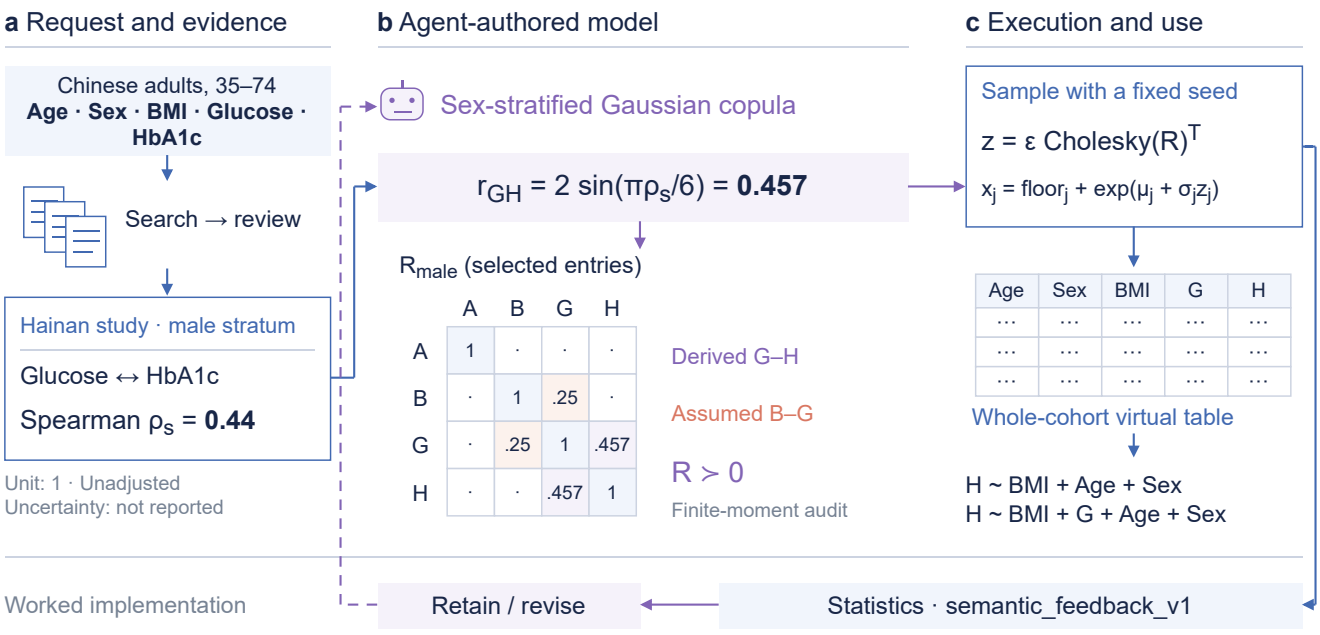


Figure 1: A completed evidence-to-cohort construction. The agent converts the Hainan male glucose–HbA1c Spearman coefficient 0.44 into latent Gaussian correlation 0.457. Selected entries of the sex-stratified copula matrix distinguish the derived glucose–HbA1c relation from the assumed BMI–glucose value 0.25. Standard-normal noise, Cholesky sampling and shifted-lognormal transforms generate BMI, glucose and HbA1c; age has a bounded-integer transform. Table cells are schematic. The scientific law, assumptions, execution checks and requested regressions form one inspectable path. This worked case is separate from the information-contrast and cross-task experiments below.

structure, so the specification records how those properties were supplied. Reviews inspect source meaning, numerical derivations, specification/code consistency and required finite moments before execution.

Figure 1 presents a completed construction using a sex-stratified Gaussian copula and positive biomarker margins. The agent converted a reported glucose–HbA1c Spearman relationship into a latent Gaussian correlation, declared the missing BMI–glucose dependence and implemented sampling and measurement transforms. Other requests led to bounded conditional laws, gamma innovations, lognormal mixtures or a recursive longitudinal family. The scientific contribution lies in choosing and connecting these components to the evidence, rather than introducing a new distribution family.

3.4 Execute and inspect the resulting analyses

Deterministic tools execute the authored program through an isolated WSL/bubblewrap adapter and return allowed source and generated-data diagnostics. Semantic checks retain the declared measurement, operator and role; unsupported comparisons remain unavailable. The author records whether to retain or revise the model within the task budget. Candidate versions, final seeds and evaluation contracts are fixed before the corresponding execution or comparison.

Researchers receive the complete tables and their scientific specification, then apply ordinary regression, correlation or longitudinal-analysis software. This separates model authorship from reproducible numerical execution while keeping the analysis request central to both. Independent numerical implementations check final statistics and reference mappings. The workflow reduces the need for researchers to hand-author each generator equation, although human preparation-time savings were not measured.

4 Evaluation design

For each request we fixed the scientific analysis and ran it on both the generated records and the appropriate reference (Table 1). When no target observations were supplied, the generator sampled every requested column. A construction is a separately authored probability model; a seed samples that fixed model. Multiple seeds measure sampling variation under its assumptions, not uncertainty about whether the assumptions hold. We report effects in their scientific units and preserve each case's original decision rule.

CHNS (6,631 adults, ages 35–74) uses unweighted estimates for the selected sample. ENSANUT (6,931 adults, ages 25–79) assigns equal mass to eight sex-by-age cells. In Q1, each case and information condition has five fresh agent authors and one prescribed statistical law. We average absolute discrepancy over three draws within each law, then across laws within the arm. E0 and E1 authors

Table 1: What the constructor could know, and what the evaluation can establish. Q1–Q5 are the identifiers of the frozen experiments. The iron study and the earlier prospective tests have separate protocols. All listed tasks generate complete cohorts.

Study	Construction inputs	Constructions and final deliveries	Analysis
Q1: CHNS and ENSANUT	Basic evidence (E0); E0 plus external relations (E1)	20 agent + 4 prescribed; 72 tables of 3,000	Six adjusted glycemia coefficients
Iron status	External evidence; target eligibility; no target biomarkers	2 agent; 6 tables of 5,000	Threshold association, joint burden and continuous ranks
Q5: glycemia and lipids	Reviewed external papers, population description, optional retrieval	6 agent; 18 tables of 3,000	Six requested effects or correlations
Earlier prospective wave	External summaries and population description; exact target results withheld	5 agent + 1 prescribed per case; 36 tables of 5,000	HAALSI glucose and Turkana lipid analyses
Q3: longitudinal reconstruction	Published HRS fitted-model artifacts	1 agent + 1 reconstructed; 6 valid tables, 11,509 subjects × 4 visits each	Age–HbA1c interactions in memory trajectories
Q2 / Q4: supporting conditions	Q2: with or without 30 target summaries; Q4: open paper retrieval without bibliographic seeds	Q2: 6 agent laws / 18 distinct tables; Q4: 4 / 12	Reconstruction of supplied summaries; repeated open retrieval

are not paired. The prescribed law illustrates a statistical completion from the same information; it is not a competitive search over statistical methods.

The reference conditions differ. Q1–Q5 reused project-level references where applicable. For iron status we sealed the protocol and the six deliveries before acquiring the target biomarkers; NHANES was already known to the project. In the earlier prospective wave, a custodian withheld the exact HAALSI and Turkana results, although investigators knew related Turkana findings. We distinguish target sampling intervals, resampling stability bands and model-conditional intervals in the figures. None makes a sample estimate the true population value.

5 Results

5.1 RQ1: What does adding external relationship evidence change?

Across CHNS and ENSANUT, the agent arms given external relationships (E1) had smaller mean absolute discrepancies than the basic-evidence arms (E0) for five of six coefficients (Figure 2). The clearest change was the CHNS glucose slope: error fell from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. In contrast, CHNS BMI error rose from 0.020380 to 0.031557 percentage points per kg/m². All three ENSANUT discrepancies decreased. The same-information statistical construction was closer on some E1 endpoints, including CHNS glucose. The result identifies analyses for which additional evidence helped in these cases; it does not establish that agent authors outperform statistical generators.

5.2 RQ2: Which uses survive in the same iron-status cohort?

Two agents independently constructed joint models of continuous hemoglobin and ferritin. Zhu et al. studied 1,671 parous, nonlactating IUD users in northern China, two to four years postpartum [33]. Their threshold counts and ferritin geometric means by hemoglobin band provided numerical constraints. A Cuban study reported an association for inflammation-corrected ferritin [34]; both agents retained it as context rather than imposing its odds ratio on raw ferritin. The intended target was US nonpregnant women aged 20–44. No source established an assay conversion or described how to transfer the full joint distribution to that target.

The agents made different decisions where the evidence ran out. A downweighted low-ferritin records and partly decoupled hemoglobin when moving to the target. B preserved the source proportions and modeled ranks within hemoglobin bands using a copula. The host supplied neither choice. In the source setting, their final simulated odds ratios were 6.58 and 6.73, close to 6.65 from the published counts. Source review also led A to replace an upper-tail rule with infinite variance; this mathematical repair preserved thresholds and ranks and does not show that revision improved the three analysis results.

We evaluated all six sealed tables against 979 women with both measurements in the positive-phlebotomy-weight NHANES frame [35]. The primary analysis was the log odds ratio; Figure 3 reports it on the odds-ratio scale. All six estimates (A: 4.835–4.866; B: 6.011–6.565) fell below 9.28, the lower confidence limit for the survey odds ratio of 17.79 (95% CI 9.28–34.11). The two secondary analyses tell different stories. For the proportion with both values low, A yielded 6.66–6.88%, near the survey estimate of 7.63%; B yielded 10.02–10.44%. For continuous Spearman correlation, B yielded 0.391–0.430, near the measured 0.398; A yielded 0.156–0.195. No source gave that correlation, although band-specific ferritin summaries constrained it indirectly. All three estimates within a delivery used exactly the same rows; neither law was selected retrospectively.

5.3 RQ3: Which findings carry across requests and reference conditions?

For glycemia and lipids, three independent authors per request generated 18 tables in total. All 54 primary estimates agreed in point direction with their corresponding sample estimates (Figure

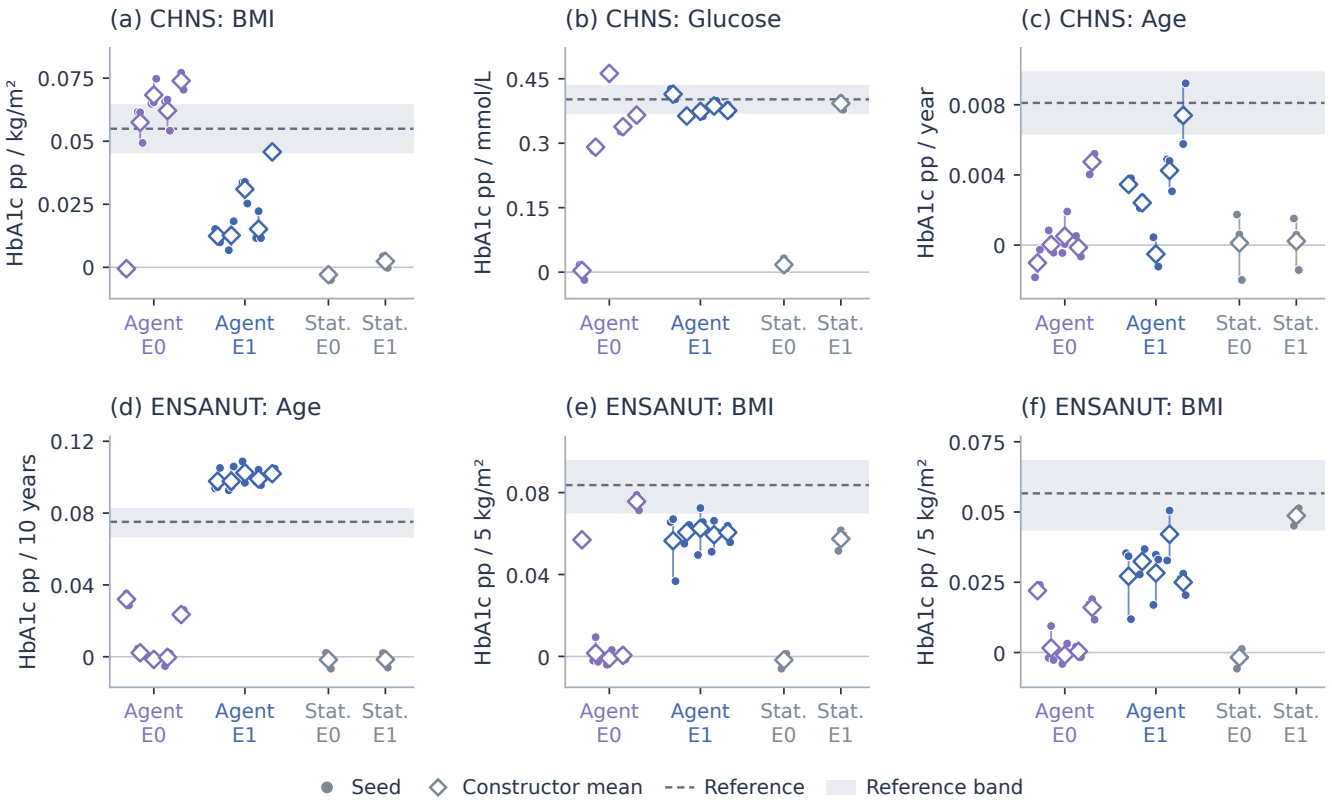


Figure 2: Relational evidence and scientific analysis agreement. CHNS panels show BMI adjusted for age/sex (a), glucose adjusted for age/sex (b), and age adjusted for sex/BMI/glucose (c). ENSANUT panels show age adjusted for sex/BMI/glucose (d), BMI adjusted for age/sex (e), and BMI additionally adjusted for glucose (f). All six analyses retain native units. E0 contains basic/marginal evidence; E1 adds relations. Each agent arm has five independently authored laws and each statistical arm one prescribed law, with all three final draws retained: 24 constructions, 72 tables and 216 coefficient estimates. Dots show seeds, open diamonds constructor means and thin segments seed ranges; dashed lines mark sample references. Reference bands are CHNS community-cluster CR1-t intervals or ENSANUT cluster-resampling stability bands. Author indices do not pair E0 with E1. Table S1 gives every endpoint's mean absolute discrepancy and E1–E0 contrast.

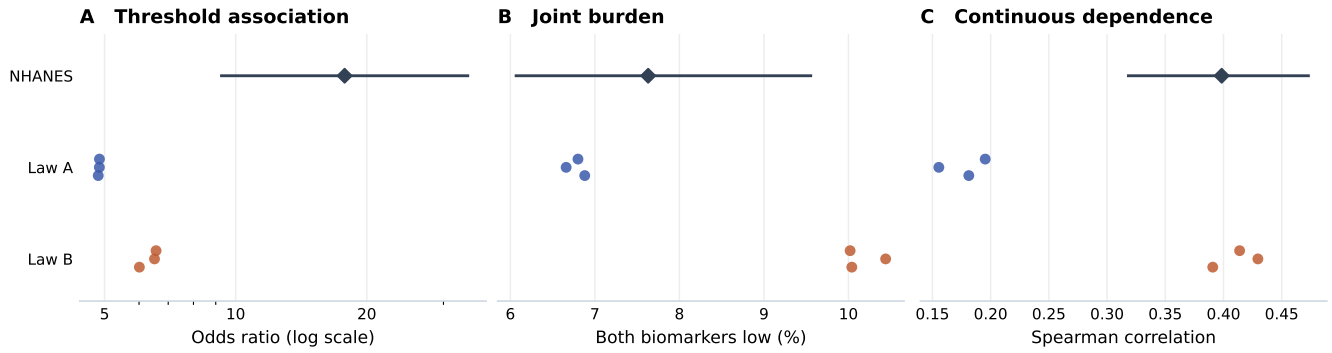


Figure 3: Matching a source association does not determine how the target records can be used. Symbols show the three draws from each frozen law and the NHANES estimate with its 95% survey-design interval. Low hemoglobin is below 12 g/dL; low raw ferritin is below 15 µg/L. Both laws understated the primary target odds ratio. The law closer for the joint proportion differed from the law closer for continuous ranks. No equivalence margin was set for these secondary analyses.

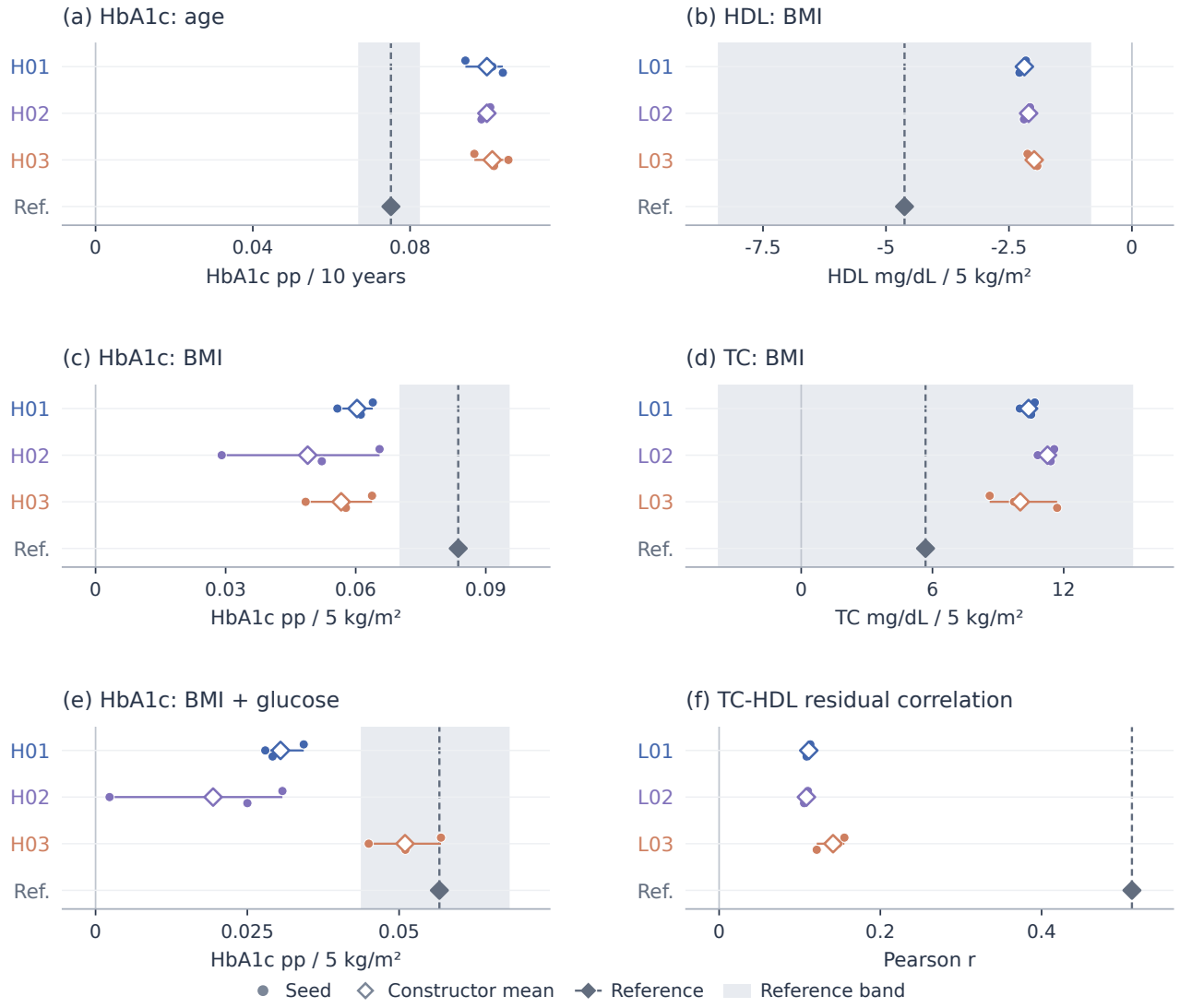


Figure 4: Analysis-specific behavior across glycemia and lipid requests. HbA1c panels in the left column show age adjusted for sex/BMI/glucose (a), BMI adjusted for age/sex (c), and BMI additionally adjusted for glucose (e). The right column shows age/sex-adjusted HDL and TC BMI slopes (b,d) and residual Pearson correlation after age/sex/BMI adjustment (f). All six laws and 54 estimates from 18 tables are retained. Circles show seeds, open diamonds means and thin segments seed ranges; dark diamonds and dashed lines mark sample references. HbA1c units are percentage points per 10 years or 5 kg/m²; lipid slopes are mg/dL per 5 kg/m². Bands are ENSANUT empirical cluster-resampling stability bands and lipid HC3 assumption-conditional sensitivity intervals. The TC band crosses zero; no residual-correlation interval is available. Seed ranges describe sampling under each frozen law. Table S5 supplies all numerical summaries and Figure S3 the dispersion results.

4). These are six different analyses, not 54 independent scientific confirmations. The tables supported age- and sex-adjusted BMI models and the further adjustment for glucose. Yet residual dependence was weaker than observed: the three constructor-mean correlations of total and HDL cholesterol after adjustment for age, sex and BMI ranged from 0.108 to 0.141, against 0.512 in the measured sample. The total-cholesterol slope's reference interval crossed zero. Thus neither matching point signs nor having plausible marginal spread established the strength of conditional lipid dependence.

The earlier prospective wave tested a stricter question. In HAALSI adults aged 40 or older [30,31], all 15 agent draws recovered the positive directions of the adjusted BMI and age associations with capillary glucose; only eight recovered the negative difference between female and male BMI slopes. The rule required four of five laws per case to meet all prespecified margins on every seed.

Three passed in HAALSI and none in Turkana [32], so the planned expansion stopped. Even a zero-effect rule met the HAALSI margin for all three endpoints. The earlier method and its gate remain a separate prospective result, not a fresh evaluation of the later workflow on the same Turkana participants.

5.4 What changes when the scientific relationship is already supplied?

In the HRS longitudinal benchmark both methods received the published model exports [1]. Below the 6.5% HbA1c knot, the age–HbA1c interaction in memory trajectories averaged −0.04464 for agent tables and −0.04050 for the reconstructed published generator, against the supplied −0.04684. Above the knot the mean signs differed; five of six intervals included zero (Figure S6). This is a matched reconstruction check, not transfer from partial evidence. Similarly, supplying 30 target summaries in Q2 allowed joint reconstruction of the five highlighted regression and correlation analyses (Table S2). In Q4, open retrieval yielded twelve distinct original studies across four authors and twelve complete tables (Tables S3–S4); the design did not isolate the effect of retrieval. HRS and Q2 test reconstruction when target-model information is supplied; Q4 instead demonstrates a way to obtain external evidence without supplying more information about the target.

6 Discussion

The product of this workflow is an analyzable virtual cohort. Its strongest property is that a researcher can follow a scientific estimate back to what the literature reported and what the agent assumed. For example, separate published projections can constrain a joint regression through a derivation (Section 3.2). A source threshold table can constrain co-occurrence without fixing continuous ranks in a different population (Figure 3). When several plausible completions yield different answers, the useful next observation is the one that distinguishes those completions: paired measurements within hemoglobin bands for the iron example, or adjusted joint lipid measurements for the residual-correlation example. Virtual records make these questions possible to formulate and test; they add no observations about actual participants.

7 Limitations and Ethical Considerations

The evidence cannot identify every joint relationship. Sources differ in eligibility, age, geography, specimen handling and adjustment; reference samples have sampling uncertainty. A small number of agent constructions share a model service, and repeated draws do not represent independent clinical cohorts. The protocols differ across studies. No held-out trial shows that review improves outcomes or that the workflow saves a measured amount of human time. For research planning, the source-based and assumed parts should travel with the records. Values must be labeled synthetic and must not be presented as measured individual biomarkers, clinical diagnoses or substitutes for independent patient evidence. Original cohort access and reuse conditions continue to apply; the public artifacts contain code and aggregate results rather than participant-level tables.

8 Generative AI Usage

Language models participated in research planning and manuscript editing and were an object of the experiments. In the scientific workflow, agent authors interpreted evidence, selected or received studies according to protocol, proposed statistical models, and wrote or revised generator code. Deterministic tools sampled the records and computed the reported analyses; separate implementations checked the numerical results where specified. Study protocols and source records document these roles. The final author list, study-specific declarations, and applicable oversight and data-use statements must be confirmed before submission.

Supporting materials and reproducibility

The public research repository contains protocols, executable code and aggregate audits. The iron protocol seal and delivery seal precede target evaluation. Supplements retain all prespecified outcomes, unfavorable cases and implementation incidents. No new cohort was generated for this revision.

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Supplementary results

The supplement retains the complete numerical comparisons and supporting displays under their original study identities. Figure and table numbers follow their new placement; no construction, primary result or unavailable comparison is removed by the shorter main narrative. Q1–Q5 remain distinct experiments with their original information conditions.

S1. Complete external-evidence contrast

All 24 Q1 constructions produced their three scheduled final tables. The 72 final tables were distinct and supported every requested regression and diagnostic; seven exploratory tables brought physical executions to 79. Additional relational evidence reduced constructor-averaged mean absolute discrepancy for five of the six designated coefficients (Table S1). CHNS glucose discrepancy decreased from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. All three ENSANUT contrasts improved. CHNS BMI discrepancy increased from 0.020380 to 0.031557.

The effect displays explain what these changes mean. CHNS E1 glucose constructor means were 0.363363–0.414189 around the 0.402093 reference. Its BMI means were positive but lower, 0.012400–0.045757 versus 0.054946. Conditional-age means ranged from −0.000509 to 0.007393 versus 0.008107; one author retained a negative mean and two negative seeds. All ENSANUT E1 focal coefficients were positive, although its age means of 0.097769–0.102452 per decade exceeded 0.075132 and its BMI means remained below their references. Preserving a direction and recovering its magnitude are distinct aspects of usefulness.

Every E1 author used at least one relationship as a numerical constraint. The complete uptake records include constraints, unfitted analytic checks and contextual uses, including the projection calculation in Section 3.2. Statistical references also achieved useful results: under E1, the prescribed constructions had lower seed-averaged discrepancies than every corresponding agent for CHNS glucose and ENSANUT conditional BMI. Other comparisons favored agent constructions. The evidence therefore supports scientific utility and the value of additional information, rather than a universal algorithmic ranking.

S2. Complete supplied-summary reconstruction

The separate CHNS experiment gave three authors external literature (L) and three authors that literature plus 30 target summaries and their metadata (R). Each author delivered three tables and had one additional diagnostic-review opportunity. All six retained their models. There were 42 physical executions: six trials and 36 initial/final deliveries. Final bytes duplicated their matching initial outputs, leaving 24 distinct tables overall and 18 distinct delivery tables. The identical stages are displayed once.

R supported close reconstruction of all five highlighted scientific analyses (Table S2; Figure S1). These values were legitimate construction inputs; their agreement measures reconstruction rather than prediction of hidden coefficients. The L/R packet and authors differ from Q1, so those two information comparisons are kept separate.

Agreement extended beyond central values. R constructor-mean BMI SDs were 3.38965–3.40243 versus 3.40632, glucose SDs were 1.48817–1.58968 versus 1.54455, and HbA1c SDs were 0.95208–0.96554 versus 0.95357. A separate three-table statistical calibration supplement also closely reconstructed the regression summaries, with mean coefficients 0.055726, 0.412568 and 0.007679. That purpose-built family was developed with coding-agent assistance and is not an agent-removal ablation. Both approaches demonstrated the usefulness of constructing a joint table from accessible aggregate information.

S3. Open retrieval across population requests

The unified workflow allocated two fresh scientific authors to each of the existing CHNS and ENSANUT requests. Each author received the intended analyses, population/measurement definitions, task interface and retrieval budget, but no target coefficients, desired signs, individual records or prior construction. CHNS- and ENSANUT-derived studies were excluded as generating evidence. Across 24 query credits, the authors selected sixteen paper uses representing twelve distinct originals. Independent review checked the original statistical definitions, numerical derivations, source applicability and required finite moments before execution. Sources could overlap across authors; four constructions do not imply four independent bodies of clinical evidence.

The agents chose materially different laws. C01 used an age-conditioned multivariate-lognormal mixture informed by Chinese population summaries [7,16] and contextual age/body-size findings [10,17]. C02 combined copula predictors with an affine HbA1c conditional mean using Pinggu summaries and Singapore coefficients [18,19], with other populations as applicability context [14,20]. E01 used bounded beta predictors and a conditional gamma response [10,17,21,22]. E02 combined lognormal BMI, conditional truncated-normal glucose and a positive conditional-mean lognormal response [10,21,23,24]. Their unreported dependencies, population transfers and residual choices remained declared assumptions. The shared controller did not prescribe these families or insert target coefficients.

Each author completed one 3,000-row trial, inspected generated-only diagnostics and source feedback, then selected its final model before three fixed draws. All four retained the initial law. The sixteen physical executions comprised four trials and twelve unique final tables, or 36,000 final virtual rows. Feedback contained 29 qualified approximate comparisons and 39 unavailable checks across the four trials, not source-population replays or 29 passed constraints. After all sessions closed, the evaluator reused the fixed sample references. A separate numerical implementation reproduced all 366 final coefficient/diagnostic values; no clinical data were downloaded or refitted.

All 36 primary seed estimates had the same positive directions as their corresponding references (Figure S2). These are repeated measurements of six analysis definitions across four laws, not 36 independent confirmations. Magnitude recovery varied by analysis. CHNS C01/C02 means were 0.03992/0.05338 for BMI against 0.05495, 0.37754/0.49241 for glucose against 0.40209, and 0.00311/0.00563 for age against 0.00811. Units are respectively HbA1c percentage points per kg/m², mmol/L and year. C02 was closer for BMI and age, while C01 was closer for glucose; no construction dominated all analyses.

Table S1: Complete main-study information contrast. Discrepancies retain the units of each row; they are not pooled. The reference response is HbA1c percentage points. E1–E0 uses unrounded quantities before display rounding.

Case / focal analysis	Predictor increment	E0 mean absolute discrepancy	E1 mean absolute discrepancy	E1 − E0
CHNS BMI, adjusted age/sex	1 kg/m²	0.020380	0.031557	+0.011178
CHNS glucose, adjusted age/sex	1 mmol/L	0.134019	0.024079	−0.109939
CHNS age, adjusted sex/BMI/glucose	1 year	0.007281	0.004854	−0.002426
ENSANUT age, adjusted sex/BMI/glucose	10 years	0.063944	0.024670	−0.039274
ENSANUT BMI, adjusted age/sex	5 kg/m²	0.056883	0.023732	−0.033151
ENSANUT BMI, adjusted age/sex/glucose	5 kg/m²	0.048797	0.025649	−0.023147

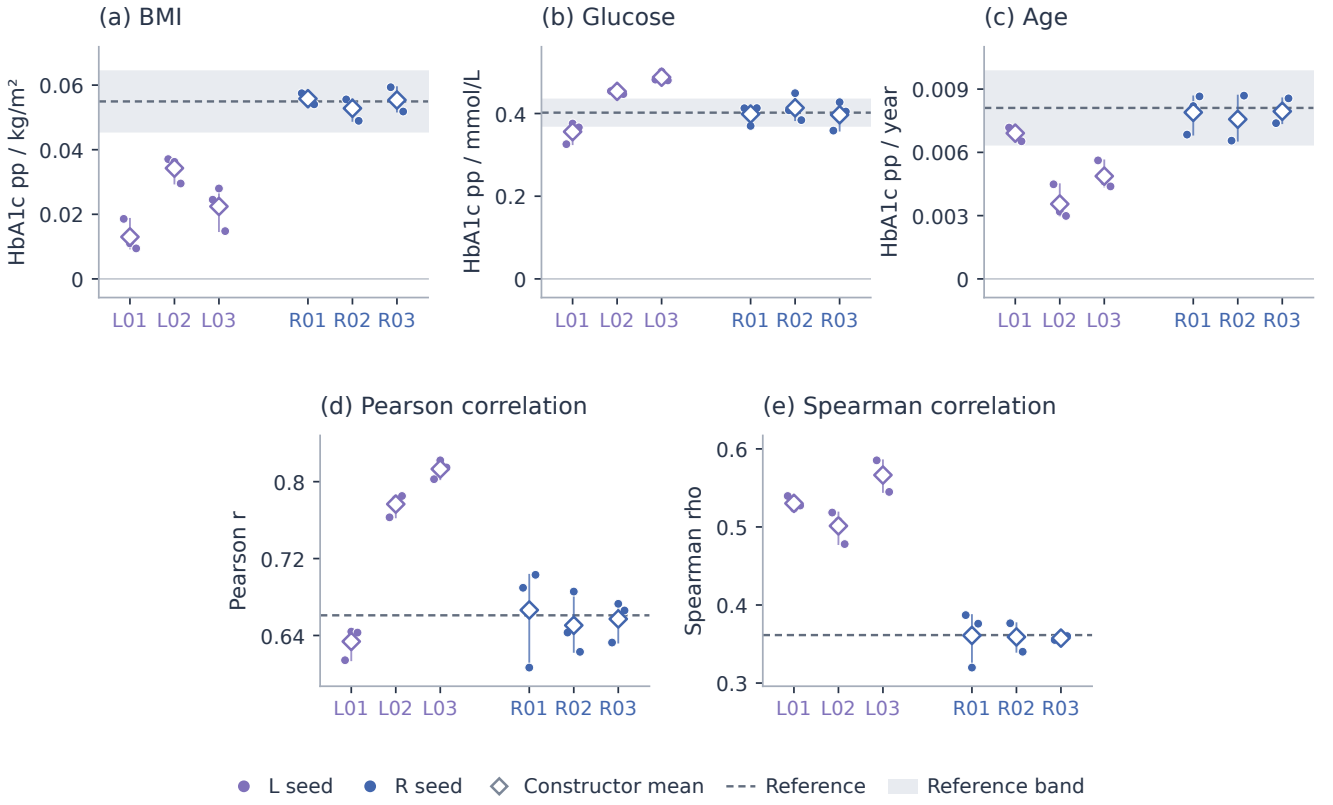


Figure S1: Supplied summaries become jointly analyzable data. Five panels show BMI, glucose and age regression coefficients plus glucose–HbA1c Pearson and Spearman correlations. L01–L03 use literature; R01–R03 additionally receive 30 target summaries and metadata. Points show all three final draws, diamonds constructor means and segments seed ranges. Dashed lines mark reference summaries available to R; regression bands are CHNS community-cluster CR1-t intervals, with no supplied correlation interval. All six constructors, 18 distinct delivery tables, 30 means and 90 seed statistics appear once. Initial and final deliveries were identical after retention; Table S2 gives the complete five-analysis numerical comparison.

Table S2: Joint-statistic reconstruction with available summaries. Ranges span three constructor means over three fixed seeds, not confidence intervals. Regression units are HbA1c percentage points per kg/m², mmol/L and year, respectively.

Analysis	Supplied reference	External literature L	Literature + summaries R
BMI coefficient, age/sex adjusted	0.054946	0.013001–0.034287	0.052821–0.055789
Glucose coefficient, age/sex adjusted	0.402093	0.356064–0.487815	0.396941–0.413715
Age coefficient, sex/BMI/glucose adjusted	0.008107	0.003552–0.006910	0.007569–0.007947
Glucose–HbA1c Pearson correlation	0.660968	0.633764–0.813170	0.650561–0.666438
Glucose–HbA1c Spearman correlation	0.361457	0.501339–0.566477	0.357571–0.360983

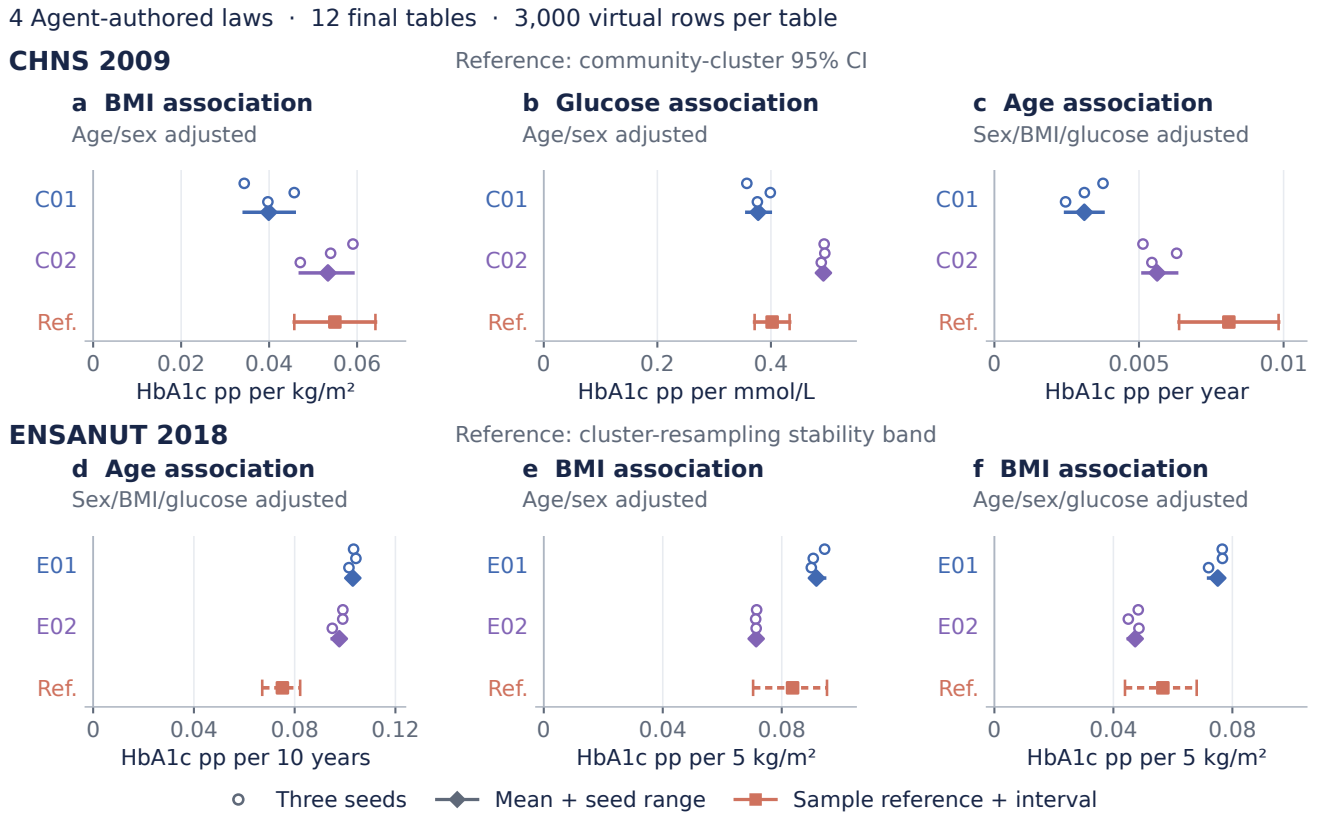


Figure S2: Repeated open-retrieval construction across two biomedical requests. All six prespecified primary analyses are shown for four agent-authored laws and twelve final 3,000-row virtual tables. Blue and purple identify the two constructions within each request. Open circles show seeds 8101–8103, ordered top to bottom within each author row; vertical spacing is solely for legibility. Diamonds and horizontal segments below the circles show the three-seed mean and minimum–maximum range. Coral squares in separate reference rows show previously exposed selected-sample estimates. Solid coral intervals are CHNS community-cluster CR1-t 95% confidence intervals; dashed coral intervals are ENSANUT empirical cluster-resampling stability bands from 399 attempts. These reference intervals have different meanings and neither is an acceptance or equivalence threshold. Axes retain native effect units and zero; pp denotes HbA1c percentage points. All generated coefficients are positive, while both CHNS age estimates are weaker and both ENSANUT age estimates stronger than their references. Seed ranges describe sampling under each fixed law, not confidence intervals or uncertainty over evidence, model choice or transport. The four constructions provide developmental workflow replication on two existing references, not independent clinical replication.

For ENSANUT, E01/E02 means were 0.10308/0.09772 for adjusted age against 0.07513 pp/decade, 0.09165/0.07140 for age/sex-adjusted BMI against 0.08364 pp per 5 kg/m², and 0.07509/0.04733 after additionally adjusting for glucose against 0.05664. Every draw retained the reduction of the BMI coefficient after glucose adjustment, an association pattern rather than causal mediation. Both authors built an age coefficient near 0.10 pp/decade and shared some sources; their agreement does not establish independent evidence convergence.

The complete diagnostics reveal remaining limits. CHNS C01 glucose–HbA1c Pearson correlation was approximately 0.666 versus 0.661, but Spearman correlation was 0.635 versus 0.361. ENSANUT residual HbA1c RMS was approximately 0.295/0.324 versus 0.488, and total SD was 0.362/0.406 versus 0.527. Correct directions and a close Pearson coefficient therefore coexisted with rank and dispersion mismatches. The complete replication report, every diagnostic and Supplementary Table S3 retain the full results. A scalar-type interface defect in the original evaluator was corrected additively on unchanged frozen tables, with its failed output and independent audit preserved; no scientific model was adjusted after comparison.

CHNS BMI/glucose analyses adjust for age and sex; CHNS age adjusts for sex, BMI and glucose. ENSANUT age adjusts for sex, BMI and glucose; its BMI analyses adjust for age/sex, with or without glucose. All Q4 diagnostic values remain in the complete diagnostic table.

The reviewed evidence/specification files distinguish reported statistics, algebraic derivations and assumed relations. The public experiment record includes per-source eligibility, measurement, adjustment, overlap and reporting-quality notes; full third-party originals and participant-like rows remain local.

S4. Longitudinal reconstruction from published artifacts

The HRS recipient-stage example [1] tested a different data product. A fresh agent received published model exports, a variable specification and the intended mixed-model analysis. It authored the recursive conditional generating family with parameter provenance. A separate implementation reconstructed the published law. Each generated 11,509 subjects at 0, 2, 4 and 6 years, producing 46,036 rows per table. No HRS individual records were used, and the source analysis coefficients were construction inputs.

All three agent and three corrected-law tables supported the native R/lme4 model with 16 fixed effects, individual intercepts/slopes and piecewise HbA1c terms around 6.5%, with age centered at 60 and scaled per decade. The below-threshold HbA1c-by-age interaction averaged −0.044635 in agent data, with range −0.051059 to −0.038247, against the supplied −0.046837. The reconstructed law averaged −0.040500. Both preserved its negative direction in all three draws (Figure S6).

Figure S6 shows the estimates and their per-table Wald intervals. The longitudinal analysis record gives all 16 fixed effects and the comparator correction.

The weaker above-threshold interaction was less stable: agent estimates averaged 0.026485 against 0.012867, while corrected-law estimates averaged −0.011691. Overall agent HbA1c mean/SD was 5.8725/0.9676 versus the original published 5.9/1.0. The high-HbA1c subgroup had mean/SD 7.1515/0.5630, compared with original 7.7/1.4 and published synthetic 7.2/0.6. Thus useful source-model behavior coexisted with inadequate upper-group spread.

A CRLF-sensitive parser initially overwrote seven coefficient/SE pairs in our comparator preparation. A separately reviewed source-bound repair produced the three corrected comparator tables without changing the frozen agent outputs. The full record contains nine physical generations, six valid comparison tables and three invalid initial comparator attempts. This was our implementation defect, not a failure of the published method. The incident record and complete effects retain the correction and all outcomes.

S5. Cross-task construction, dispersion and source uses

The next study applied the common workflow to HbA1c analyses in the ENSANUT2018 analytical design and BMI-related TC/HDL analyses in the Turkana study context. Three fresh agent contexts constructed a law for each request, using reviewed external articles, population/measurement metadata and optional literature tools. No participant inputs, target coefficients, target means or prior generators were supplied. The six contexts used the same model service and shared some source evidence. This study therefore tests repeated construction across two requests, with the law chosen by each author, rather than independent clinical replication or a controlled comparison of workflow components.

The HbA1c analyses retained the existing age and BMI estimands and equal allocation to eight sex-by-age cells. The lipid analyses used unweighted age/sex-adjusted BMI coefficients for TC and HDL, plus the Pearson correlation of their residuals after age, sex and BMI adjustment. Lipid eligibility required adults with recognized recorded sex and finite, positive BMI and lipid values. Generated lipid concentrations were converted from mmol/L to mg/dL using the frozen factor 38.67. The reference represents a selected Turkana/Laikipia study sample; fasting, assay transfer and population composition remained qualified in the evidence record. No new participant records were read, downloaded or refitted for this study.

The agents made different scientific choices within the common interface. H01 combined bounded predictors, conditionally truncated glucose and a shifted gamma HbA1c innovation; H02 used bounded beta conditional laws. H03 solved cross-source moment equations under declared dependence and dispersion assumptions, then used conditional beta laws. Their source records included overlapping age, body-size and glycemia evidence [10,17,21,22], with additional check-only/contextual evidence for H01 [14]. The lipid authors combined Gaussian copulas and lognormal margins using geometric summaries from Greiner et al. [25], rural Ashanti rank correlations [26] and arithmetic moments from Agongo et al. [27,28]. Choices about arithmetic versus geometric location, transfer of coefficients of variation, sex strata and age dependence differed among L01–L03. These are applications of established probability families whose numerical use was selected and explained by the agent.

Independent source and implementation reviews checked the statistical meanings, specification/code consistency and finite moments before isolated execution. Each constructor completed one

Table S3: Complete Q4 open-retrieval primary summaries. Means and ranges summarize three final seeds under each fixed construction. Units are HbA1c percentage points per stated predictor increment. No row is omitted; ranges are not confidence intervals.

Request / analysis	Author	Increment	Reference	Generated mean	Seed min–max
CHNS BMI	C01	1 kg/m²	0.054946	0.039923	0.034342–0.045708
CHNS glucose	C01	1 mmol/L	0.402093	0.377538	0.357560–0.398892
CHNS adjusted age	C01	1 year	0.008107	0.003109	0.002461–0.003755
CHNS BMI	C02	1 kg/m²	0.054946	0.053375	0.047088–0.059055
CHNS glucose	C02	1 mmol/L	0.402093	0.492412	0.488649–0.494875
CHNS adjusted age	C02	1 year	0.008107	0.005628	0.005137–0.006301
ENSANUT adjusted age	E01	10 years	0.075132	0.103084	0.101546–0.104315
ENSANUT BMI	E01	5 kg/m²	0.083643	0.091651	0.089940–0.094413
ENSANUT glucose-adjusted BMI	E01	5 kg/m²	0.056638	0.075089	0.072005–0.076661
ENSANUT adjusted age	E02	10 years	0.075132	0.097724	0.094947–0.099135
ENSANUT BMI	E02	5 kg/m²	0.083643	0.071399	0.071232–0.071563
ENSANUT glucose-adjusted BMI	E02	5 kg/m²	0.056638	0.047325	0.045022–0.048592

Table S4: Original-source inventory for Q4 open retrieval. Twelve unique originals supplied sixteen author-specific paper uses. Sharing a source is not independent clinical evidence; source populations are not the target populations.

Original	Reference	Constructing authors
Wu et al., 2017	[16]; PMC5591004	C01
Bao et al., 2010	[7]; PMC2871988	C01
Pani et al., 2008	[10]; PMC2551641	C01, E01, E02
Skogberg et al., 2019	[17]; PMC6458679	C01, E01
Zhang et al., 2016	[18]; PMC5078665	C02
Venkataraman et al., 2012	[19]; PMC3504343	C02
Liu et al., 2016	[20]; PMC5009131	C02
Wisgerhof et al., 2020	[14]; PMC7272077	C02
Huang et al., 2021	[21]; PMC8038122	E01, E02
Ghazanfari et al., 2010	[22]; PMC3075530	E01
Masuch et al., 2019	[23]; PMC6371438	E02
Marini et al., 2017	[24]; PMC5398507	E02

scheduled trial, inspected generated diagnostics and source feedback, and could revise once. All six retained their initial laws. Three fixed final seeds, 9101–9103, produced 18 distinct 3,000-row tables: 54,000 fictional rows from 24 successful physical attempts, including the six trials. No final execution failed and no seed was replaced. Reference comparison began after all six author sessions closed and reused previously exposed aggregate results.

All 54 primary point estimates had the same direction as their corresponding sample estimates, with quantitative agreement varying by analysis (Figure 4; Table S5). HbA1c age effects averaged 0.09954–0.10091 percentage points per decade versus 0.07513 in the reference; age/sex-adjusted BMI effects averaged 0.04892–0.06030 versus 0.08364 per 5 kg/m². With glucose additionally adjusted, H03 averaged 0.05098 versus 0.05664, whereas H01 and H02 averaged 0.03047 and 0.01936. Thus the shared age direction and attenuation after glucose adjustment did not imply recovery of each effect magnitude.

For lipids, mean HDL BMI coefficients ranged from −2.1865 to −1.9868 mg/dL per 5 kg/m² versus −4.6230 in the reference, while TC coefficients ranged from 10.0151 to 11.2609 versus 5.6831. The TC reference sensitivity interval was −3.7535 to 15.1197, so matching its positive point estimate does not establish replication of a definite positive population effect. Residual TC–HDL correlations averaged 0.10828–0.14134 versus 0.51207. The generators preserved the directions of these focal point estimates while substantially underrepresenting the lipid dependence remaining after the specified adjustment.

Dispersion changes the assessment of usefulness (Figure S3). H03's closer glucose-adjusted BMI coefficient accompanied lower

HbA1c marginal SD and full-model residual RMS, 0.3523 and 0.3006 percentage points versus 0.5274 and 0.4885. H02 retained more variation, with corresponding values 0.4825 and 0.4036, but smaller BMI coefficients. L02's TC SD was close to the reference, 43.6778 versus 42.8466 mg/dL, although its residual TC–HDL correlation remained weak. All three lipid laws had HDL SDs below the reference. No construction uniformly dominated across focal effects, dependence and dispersion.

The evaluation retained all 513 prespecified estimates and diagnostics, including the 54 primary outputs. Independent recomputation verified every value and all 324 matched reference comparisons; the maximum numerical difference was 5.33×10^{-14}. The remaining 189 diagnostics lacked matching cached reference estimators and remained generated-only results. Across authors, eight original studies supplied 22 author–paper uses, including context/check-only uses; optional retrieval consumed seven query credits and one original download. All 132 declared source-check entries were unavailable under the implemented semantic rules because weighting or essential operator definitions were unresolved or unsupported (Table S6). Numerical source values still informed reviewed parameter derivations, but this batch provides neither measured numerical source-fit feedback nor a before/after revision benefit. The complete results and reviews retain every endpoint, unavailable comparison and implementation incident, including correction of the independent auditor's identifier parser without changing any generated table.

HbA1c age adjusts for sex, BMI and glucose. HbA1c BMI adjusts for age/sex, with glucose added in the third analysis. Lipid BMI slopes adjust for age/sex; the residual correlation removes age/sex/BMI projections from both lipids on the same rows. H01–H03 belong only to this Q5 HbA1c request, and L01–L03 only to this Q5 lipid request; the L labels do not refer to the earlier Q2 literature arm. All 54 individual primary outputs and all 513 prespecified estimates/diagnostics remain in the complete endpoint table. The 324 matched reference comparisons and 189 generated-only diagnostics retain their original availability labels.

The evidence-use inventory and source/code review retain the source IDs, numerical roles, unresolved definitions and actual retrieval activity. Unresolved weighting and unsupported or incompletely specified operators were not replaced with different statistics. Parameter derivations can therefore use reviewed source information while the corresponding runtime source comparison remains unavailable.

S6. Dependence sensitivity and the supporting construction

The preceding open-search example acquired five seriously reviewed studies [9,10,12,13,14] without bibliographic seeds or desired signs. A fresh constructor selected a sex-stratified Gaussian copula with positive biomarker margins and generated one trial plus three complete 3,000-person tables. It retained the model after eleven approximate and 23 unavailable source comparisons per execution. Secondary BMI coefficients were 0.04217–0.05176 versus 0.05495 in the reused CHNS reference, secondary age was 0.01613–0.01771 versus 0.01530, and primary age was 0.00842–0.01084 versus 0.00811. Primary BMI/glucose were attenuated, and the female point direction and distributional tails remained imperfect. Supplementary Figure S4 retains all seven nonintercept effects and their uncertainty labels.

Before generating data or comparing targets, the source review identified an unreported BMI–glucose dependence. Two valid latent correlations, 0.10 and 0.40, implied conditional BMI coefficients of +0.03422 and −0.01095 while retaining the same margins, other bivariate laws and secondary projection. The delivered model used the declared assumption 0.25. This is an analytical sensitivity, not additional generated cohorts or a causal intervention. It demonstrates why an explicit data-generating law can identify which missing relationship matters to an intended use.

A separate analysis contextualized the same three tables with 100 community-disjoint resplits of the existing CHNS sample, drawing 3,000 people per side and retaining all 36 metrics. Secondary BMI and age discrepancies were on scales comparable with within-sample variation, while primary glucose/female and several joint/tail discrepancies were larger. The complete table preserves the nonexchangeable real-real and generated comparisons. These are descriptive interpretation aids, not a new population or retrospectively selected equivalence cutoff.

S7. Prospective evaluation under the earlier method

In HAALSI, eligible ages started at 40 years and the analysis used the released capillary point-of-care glucose measurement. Fasting was not an eligibility requirement. The fixed BMI and age contrasts were taken from the same additive glucose model with recorded sex. A second model estimated the difference in the BMI slope between women and men, adjusting for age. In Turkana, the two BMI slopes were adjusted for age and recorded sex. The correlation uses the residuals after adjusting both lipid measurements for age, sex and BMI.

Each case had five separately authored agent laws and one prespecified statistical generator (S01). The experiment produced 48 source-context and 36 target-context tables of 5,000 rows each. All authors retained their initial laws after source-only feedback. Initial and final labels therefore refer to the same 36 target tables. The public checkpoint preceding disclosure records work in progress. The final evaluation record documents the complete inventory, one reveal and independent recomputation of 6,704 checks. HAALSI had a prospectively withheld target. Turkana's exact estimates were withheld, with prior investigator exposure to adjacent findings. Later Q5 reused the same 397-person Turkana reference.

The primary rule divided the absolute difference in each slope by the measured outcome's sample SD and used the absolute correlation difference directly. A construction passed when all nine comparisons (three analyses by three seeds) were at most 0.10. The agent counts at thresholds 0.05, 0.10, 0.20 and 0.30 were 1, 3, 5 and 5 of five for HAALSI and 0, 0, 1 and 4 for Turkana. Both S01 laws missed the primary criterion. For HAALSI, the discrepancies for three zero-valued effects were 0.0990, 0.0733 and 0.0770. The frozen decision closed the expansion. The prospective study predates the later interface for documenting numerical evidence use and supplies a separate evaluation of the underlying construction approach.

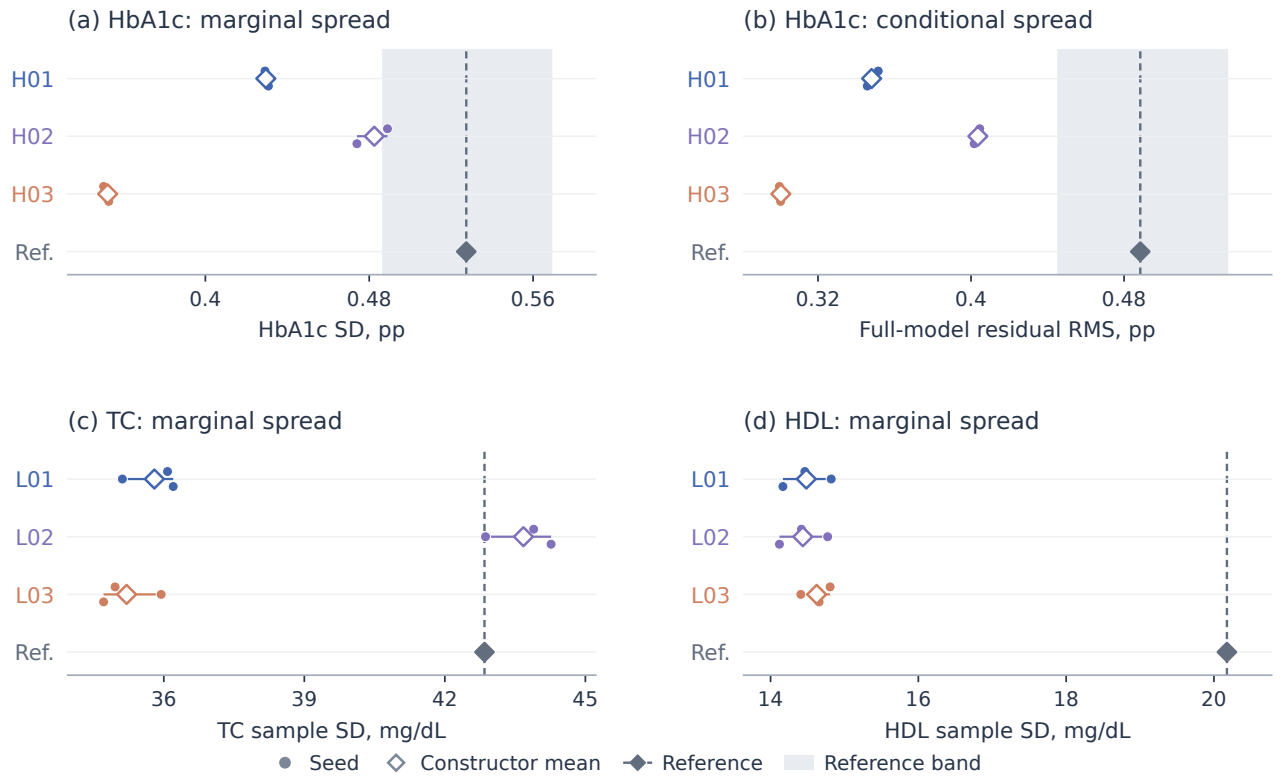


Figure S3: Dispersion across all cross-task constructions. Panels show marginal HbA1c SD, HbA1c full-model residual RMS, and marginal TC and HDL SDs for each construction and its three final draws, retaining all 36 diagnostic values from the same 18 tables. HbA1c uses the population-SD diagnostic; lipids use sample SD. These quantities measure individual or residual spread, not uncertainty in a regression coefficient. H03's relatively close adjusted BMI effect in Figure 4 coexists with reduced HbA1c spread, and L02's close TC SD coexists with weak adjusted lipid dependence. Seed ranges reflect Monte Carlo variation conditional on the selected law. Gray HbA1c bands are empirical cluster-resampling stability bands; no reference interval is supplied for lipid SDs.

S8. Full records and retained implementation incidents

The public repository provides original code, protocols, aggregate results and independent checks. The complete v0.9 extended record preserves all six original numerical tables and full methods/results. Its Table 2 supplies every main endpoint's adjustment set, units and reference interval; Tables 3–4 retain the main and summary-assisted values; Table 5 retains all latest seven effects; Table 6 retains the real-real comparison values. These archived tables preserve the earlier analyses alongside the separately sealed iron study.

The main study, summary-assisted study, statistical supplement, HRS example, preceding open-search construction, unified workflow replication and cross-task evidence-use study preserve separate frozen identities and complete endpoint outputs. The cross-task protocol, complete endpoint table, independent numerical review and scientific interpretation document the new unit. Earlier live A/B results, including the conditioned BMI mismatch and source-replay unavailability, remain in the extended record. Figure updates and added tables consume reviewed aggregates only; manuscript integration creates no cohorts or clinical fits. In the main study, seven authors inspected exploratory outputs and kept their laws, and thirteen finalized without exploration. The two earlier live authors both kept their models after the available review.

Individual real/generated tables, third-party originals and credentials remain local; clinical reruns require lawful access and the appropriate environment. Historical model identifiers and token costs are incompletely metered. Author/affiliation/funding/conflict/ethics/access/AI declarations and final submission formatting remain to be completed. No manuscript has been submitted.

The historical supplement retains earlier enrichment and direct-row construction, the divergent-moment repair, failed transport criteria, tail defects and revision trade-offs. The v0.16 reading draft preserves the earlier layout.

Table S5: Complete Q5 cross-task primary summaries. Each row summarizes three fixed final draws from one law. All 18 author–analysis combinations are retained. HbA1c entries are percentage points (pp); lipid slopes are mg/dL per 5 kg/m². Ranges are seed minima and maxima, not confidence intervals or evidence/model uncertainty.

Request / analysis	Author	Units / increment	Reference	Generated mean	Seed min–max
HbA1c adjusted age	H01	pp / 10 years	0.075132	0.099545	0.094075–0.103606
HbA1c BMI	H01	pp / 5 kg/m²	0.083643	0.060295	0.055769–0.063938
HbA1c glucose-adjusted BMI	H01	pp / 5 kg/m²	0.056638	0.030469	0.027956–0.034291
HbA1c adjusted age	H02	pp / 10 years	0.075132	0.099539	0.098162–0.100396
HbA1c BMI	H02	pp / 5 kg/m²	0.083643	0.048923	0.029090–0.065499
HbA1c glucose-adjusted BMI	H02	pp / 5 kg/m²	0.056638	0.019363	0.002308–0.030777
HbA1c adjusted age	H03	pp / 10 years	0.075132	0.100912	0.096356–0.105015
HbA1c BMI	H03	pp / 5 kg/m²	0.083643	0.056649	0.048449–0.063758
HbA1c glucose-adjusted BMI	H03	pp / 5 kg/m²	0.056638	0.050978	0.044994–0.056901
HDL BMI	L01	mg/dL / 5 kg/m²	−4.623004	−2.186526	−2.286805–−2.118624
TC BMI	L01	mg/dL / 5 kg/m²	5.683108	10.393977	9.992186–10.676486
TC–HDL residual correlation	L01	dimensionless	0.512070	0.111234	0.108517–0.113140
HDL BMI	L02	mg/dL / 5 kg/m²	−4.623004	−2.100390	−2.194806–−2.035712
TC BMI	L02	mg/dL / 5 kg/m²	5.683108	11.260864	10.811415–11.565687
TC–HDL residual correlation	L02	dimensionless	0.512070	0.108283	0.105395–0.110194
HDL BMI	L03	mg/dL / 5 kg/m²	−4.623004	−1.986781	−2.122835–−1.913541
TC BMI	L03	mg/dL / 5 kg/m²	5.683108	10.015112	8.617345–11.698752
TC–HDL residual correlation	L03	dimensionless	0.512070	0.141343	0.121087–0.155368

Table S6: Q5 original-source uses and unavailable feedback. Eight distinct originals supplied 22 author–paper uses, including context/check-only evidence. All six authors retained their initial laws. The 132 source-check entries were unavailable, not successful numerical comparisons or failed generators. The inventory counts the three lipid originals, with the Agongo correction [28] attached to its source rather than counted as a separate original study.

Author	Original-source references	Paper uses	Unavailable source checks
H01	Pani [10]; Wisgerhof [14]; Skogberg [17]; Huang [21]; Ghazanfari [22]	5	13
H02	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	18
H03	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	32
L01	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	29
L02	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	19
L03	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	21
Total	8 distinct original studies	22	132

S9. Iron-status analysis and independent verification

All six deliveries used two continuous columns, with three target analyses per delivery. The target survey comparison used the phlebotomy weight and 15 design degrees of freedom; 979 of 1,065 eligible positive-weight women had both measurements. A separate implementation reproduced all 21 point estimates (three reference, eighteen generated), the six target interval endpoints and the four unweighted cell counts. The study report gives all seed values and

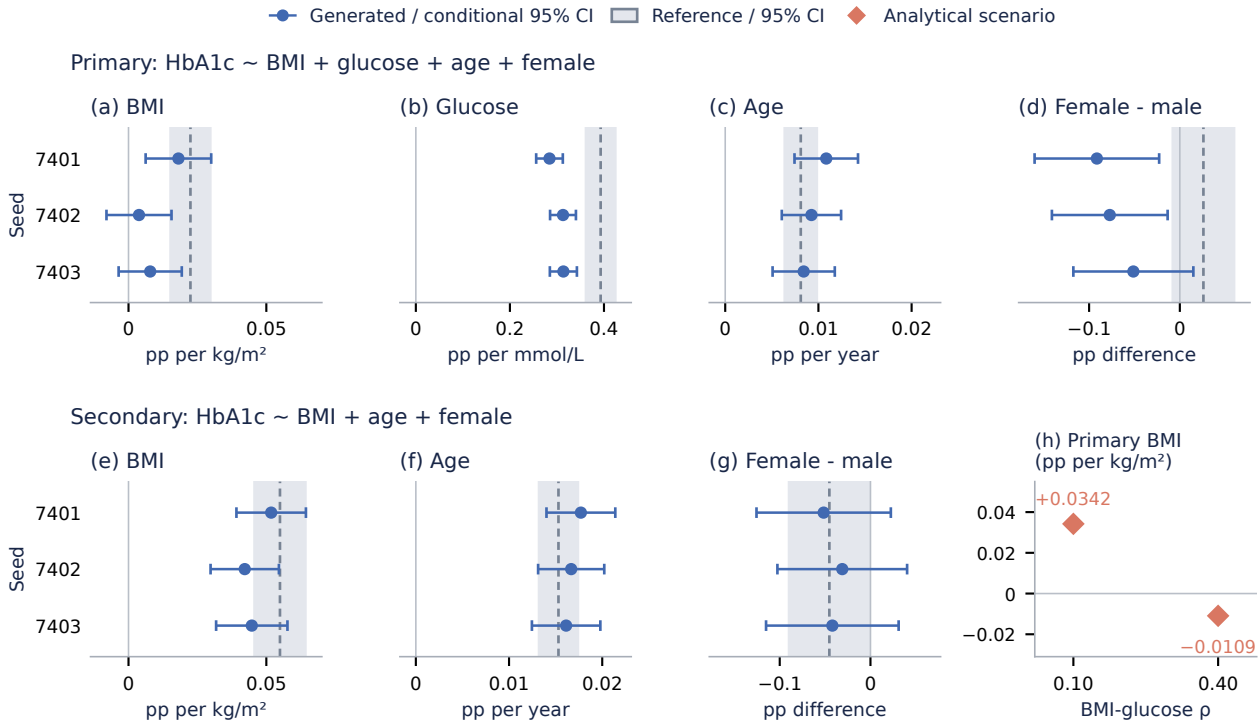


Figure S4: Analysis-specific usefulness and an unresolved joint dependence. Panels (a–d) show all primary-model nonintercept effects and (e–g) all secondary-model effects. Blue points are three fixed 3,000-person draws with conditional 95% HC3-t intervals. Gray dashed lines and bands are exposed CHNS reference estimates and 95% community-cluster CR1-t intervals; solid lines mark zero. All effects retain native units (pp: HbA1c percentage points), with shared limits for matching predictors. Generated intervals omit source, structural and transport uncertainty; overlap is not equivalence. Panel (h) displays two pre-generation analytical scenarios: latent BMI–glucose correlations 0.10 and 0.40 imply conditional BMI coefficients +0.03422 and −0.01095 while leaving marginal laws, other bivariate laws and the secondary projection unchanged. The delivered model used 0.25. These are not new draws, a fitted curve or a causal intervention. Retained initial/final stages alias the same three tables.

Table S7: Six prespecified analyses under the earlier workflow. The columns report the measured-sample estimate, the minimum and maximum across 15 agent draws, the number agreeing in point sign, and the range of the three draws of the prescribed statistical law. The ranges describe draws under the candidate models. HAALSI glucose contrasts are in mmol/L, lipid slopes in mg/dL and the adjusted correlation is unitless.

Analysis	Measured	Agent range	Same sign	S01 range
HAALSI glucose, BMI per 5 kg/m²	0.3138	0.3285 to 0.5698	15/15	−0.0174 to 0.0180
HAALSI glucose, age per 10 years	0.2322	0.1136 to 0.3005	15/15	−0.0170 to 0.0077
HAALSI female-minus-male BMI slope per 5 kg/m²	−0.2440	−0.2487 to 0.1903	8/15	−0.0524 to 0.0046
Turkana HDL, BMI per 5 kg/m²	−4.6230	−3.5509 to −1.8173	15/15	−0.1120 to 0.0710
Turkana TC, BMI per 5 kg/m²	5.6831	10.2822 to 15.0413	15/15	−0.4657 to 0.0334
Turkana residual TC–HDL correlation	0.5121	0.1162 to 0.5361	15/15	0.3625 to 0.3840

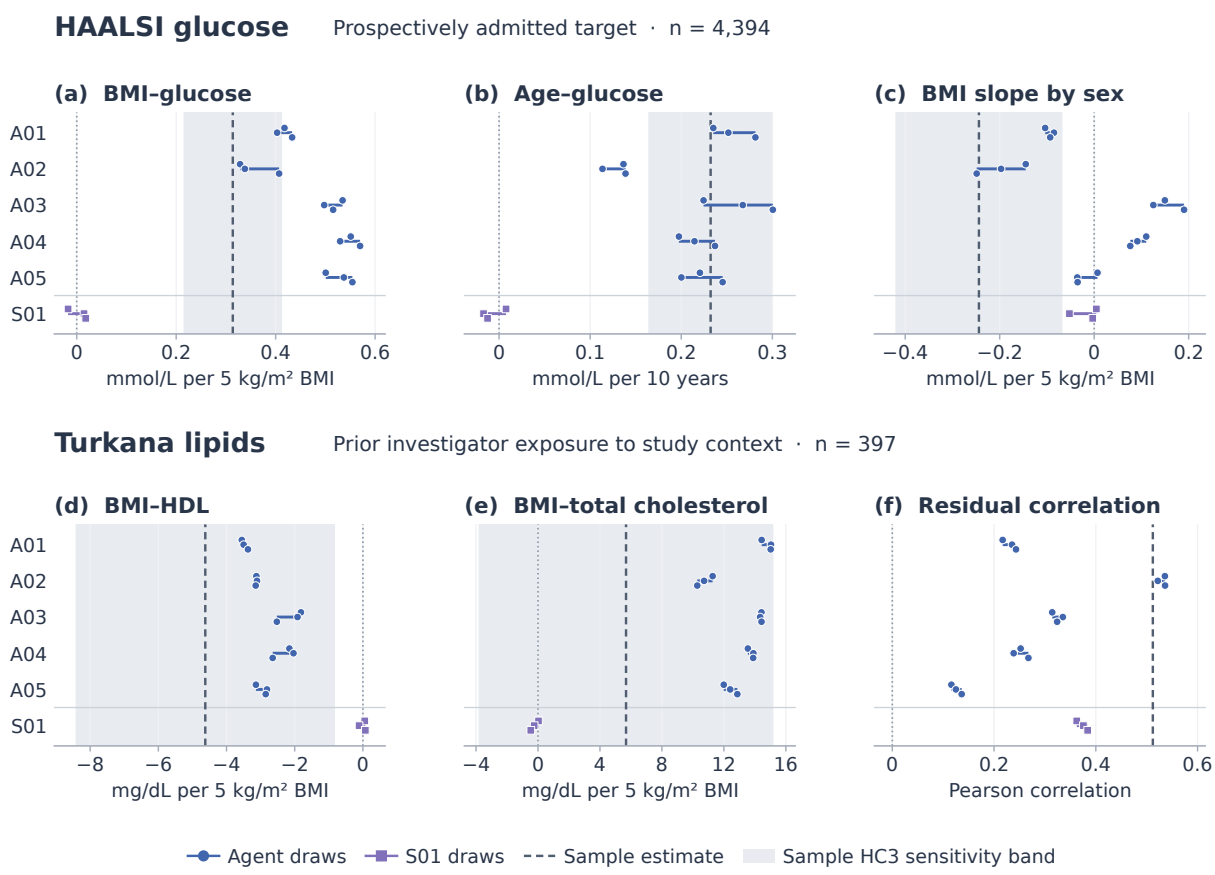


Figure S5: Earlier prospective evaluation of whole-cohort generation. Blue circles show the three draws of each agent law and purple squares the three draws of the prescribed statistical law (S01). Short segments span draws. Dashed lines mark measured-sample estimates, and shaded bands give approximate 95% HC3 sensitivity intervals for the five slopes. The residual correlation has no reference interval. HAALSI glucose main effects were directionally stable across agent draws. The sex difference in the BMI slope varied by construction. Turkana's total cholesterol point estimate had a reference interval that included zero. Its quantitative estimates were withheld before generation despite prior exposure to related published findings.

the evaluation protocol specifies the source replay, primary log-odds-ratio comparison and secondary analyses.

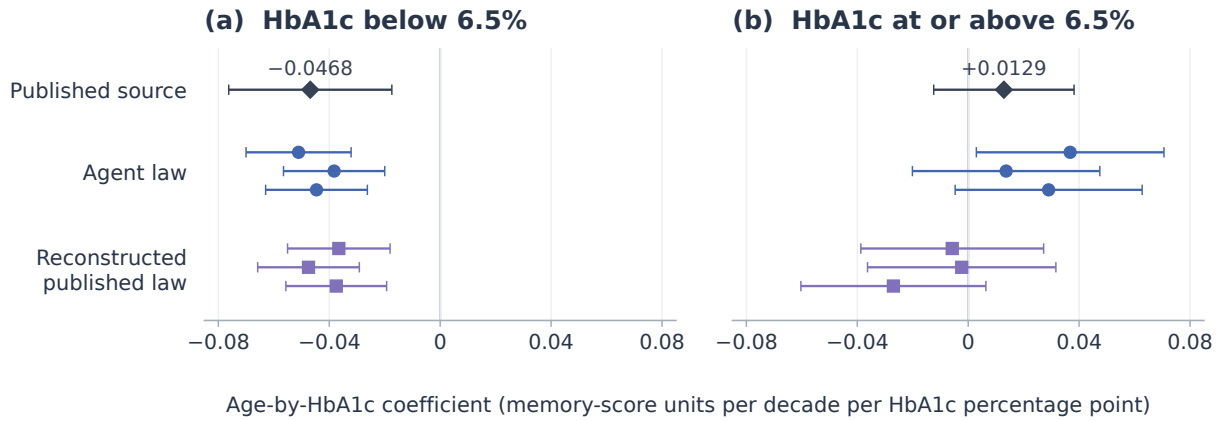


Figure S6: Two prespecified age–HbA1c interactions in a longitudinal memory model. Diamonds show the supplied coefficients; circles and squares show estimates from three draws per law (seeds 6101–6103, top to bottom). Bars show approximate 95% Wald intervals calculated from the exported source standard errors or from the generated-table fits. Units are standardized memory-score change per decade per percentage point of HbA1c within the indicated branch. Each table contains 11,509 synthetic subjects and four visits per subject. The upper-branch signs differed between constructions. The six tables describe sampling under two frozen generating laws.